

In search of the frugal innovation strategy

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Abstract

Purpose – In the past years, many contributions have been published addressing frugal innovation and other types of resource-constrained innovations in the management field. Throughout this paper frugal innovation is reported as a phenomenon, concept, research field and strategy, showing the different ways the literature refers to it and how scattered the concept is. However, based on the understanding, the authors decided to address frugal innovation as a kind of innovation strategy that helps companies to innovate in resource-constrained environments. Therefore, considering the increasing interest in the frugal innovation topic and adding the perspective of business strategy for resource-constrained conditions, the research question addressed in this paper is: what are the main features of the frugal innovation literature that unfold its current perspectives for business strategy? In this sense, the purpose of this study is to analyze the scientific production in frugal innovation through a literature mapping and review to better understand it, delimiting different perspectives and creating boundaries to other business strategies or approaches.

Design/methodology/approach – The authors performed a co-citation analysis using the Vosviewer software to notice how authors are arranged in clusters based on their understanding of the topic. Furthermore, the authors also performed a systematic literature review (SLR) analyzing the extant literature on frugal innovation based on the clusters found through the co-citation method. The final sample used in the study included 42 papers published between 2011 and 2019, using the Web of Science platform as a data source.

Findings – By means of SLR, the findings of this study provided a more organized view of frugal innovation through the co-citation analysis and the qualitative analysis of the clusters, which were the basis for the parameters established. After the conceptualization of frugal innovation strategy (FIS) and the delimitation of boundaries of FIS, the authors bring the reflections about the contribution to the literature and the practice (managers and society) by showing three assumptions to be tested and confirmed in future studies and a framework to guide companies in search of a FIS.

Originality/value – This study contributes to the discussion on frugal innovation moving a step forward to clarify the research field on this subject providing the main characteristics for researchers and practitioners. The paper has delimited the boundaries of FIS. The assumptions established in the discussion can become hypotheses for empirical studies. In addition, the authors explain why, what is, where and to whom the FIS can be developed and applied. Furthermore, the authors contribute by developing the FIS framework, with four strategic positions based on the boundaries of FIS organized by the complexity technological level. The strategic positions are frugal innovation orientation (FIO), FIO to value shared, FIO to market and FIS.

Keywords Systematic literature review, Strategy, Bibliometric mapping, Co-citation analysis, Frugal innovation

Paper type Literature review



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1. Introduction

The importance of innovation to competitiveness is acknowledged and emphasized. However, understanding the meaning of innovation seems more controversial, especially in the academic environment, where research is mostly concentrated on technological innovation when compared to other types of innovation (Birkinshaw *et al.*, 2008). Frugal innovation, for instance, has required further studies in the innovation strategy field, as several scholars and business managers started to understand its business applications and disruptive capacity (Rao, 2013).

According to Rao (2013), frugal innovation can depict a disruptive capacity that allows global companies to embrace frugality as a strategy to meet large market needs in emerging economies. Frugality is a set of principles to deal with resource constraints (Radjou *et al.*, 2013; Radjou *et al.*, 2015). In this sense, Pisoni *et al.* (2018) understand that frugal innovation has been drawing the attention of scholars and practitioners because of its potential for helping underserved markets with resource constraints, looking at them as opportunities.

Recently, many contributions have been published addressing frugal innovation and similar concepts that share some characteristics in common with other resource-constrained innovation strategies. However, scholars and practitioners often use the concepts of “cost innovation,” “good-enough innovation” and “frugal innovation” in an interchangeable manner (Zeschky *et al.*, 2014) darkening their meaning, which, in turn, hinder the understanding of the differences among them. In addition, it seems that there are misinterpretations among scholars who comprehend frugal innovation as basically low-cost design or frugal engineering and hacking, low tech or good-enough solutions (Ahuja, 2014).

In the literature, frugal innovation is reported as a phenomenon, concept, research field and strategy. It shows the different ways that the literature refers to it and how scattered the concept is. Our effort here is to organize the literature about frugal innovation and address frugal innovation as a kind of innovation strategy that helps companies to innovate in resource-constrained environments. Therefore, considering the increasing interest in the frugal innovation topic and adding the perspective of business strategy applied to resource-constrained conditions, the research question addressed in this paper is: what are the main features of the frugal innovation literature that unfold its current perspectives for business strategy? In this sense, the main purpose of the article is to delimit the boundaries of the frugal innovation strategy (FIS). After the delimitation of why, what is, where and to whom FIS can be developed and applied, we proposed a framework to guide firms in search of FIS. To do this, we analyze the scientific production in frugal innovation through a literature mapping and review to better understand the FIS. It allows specifying different perspectives, creating boundaries to other business strategies or approaches. To do so, it will be necessary to understand how frugal innovation has been addressed by its main authors and how their studies are connected to make it possible to have a better understanding. In this sense, the bibliometric mapping and the systematic literature review (SLR) are helpful to clarify similarities and distinctions and provide assumptions to better guide toward the FIS.

The main contribution of the paper is the delimitation of the boundaries of the FIS. To the literature, we explain why it can be considered as a strategy and what means a FIS, as well as where and to whom it can be developed and applied. FIS is beyond cost innovation and good-enough innovation and could be a reverse innovation but not necessarily. Also, FIS is not limited to emerging markets, it meets the specific needs of both poor and environmental-conscious consumers. Finally, FIS has an economic, social and environmental value proposition. The result of the conceptualization of FIS is the FSI to the literature. The implementation of frugal innovation requires a company strategy, therefore, we bring to the

literature four strategic positions based on the concept of FIS and an organizing framework. This framework has an impact on managers and society. For the first, to establish the firms and develop strategies in search of FIS, while for the second, the impact is to investors (public, institutional or private). This may also help companies and governments to see the benefits the frugal principles can bring to countries and markets, hence, moving forward to their economies.

2. Conceptual background

Perhaps, [Schumpeter \(1934\)](#) was one of the first authors to set different types of innovations such as innovation in products, processes of production, markets, new sources of supply and new ways of organizing any industry. However, to get a broader understanding of the meaning of innovation, it becomes important to comprehend it beyond the technological aspect ([Birkinshaw et al., 2008](#)). Frugal innovations might be considered as disrupting innovation processes that proved to be an example of Schumpeterian creative destruction, providing its potential to promote local economic development in low-income environments ([Rao, 2013](#)).

The Indian word *Jugaad*, which means “the art of overcoming harsh constraints by improvising an effective solution using limited resources,” has been considered as the precedent of the term frugal ([Radjou et al., 2012](#) in [Prabhu and Jain, 2015](#), p. 847). In this sense, *jugaad* innovation entails using creativity and improvisation to make an existing thing more efficient or to create something new with very little resources ([Ahuja, 2014](#)), both aspects of frugality. It is:

[...] commonly considered as an example or a sub-set of frugal innovation that stresses the concept of improvisation, especially related to countries like India and China, where frugality is one of the key elements along with flexibility and inclusivity ([Pisoni et al., 2018](#), p. 9).

Frugality is a set of principles that researcher Navi Radjou has established to describe “*jugaad* innovations” in poor countries ([Radjou et al., 2013](#)) and “frugal innovations” in rich countries ([Radjou et al., 2015](#)). *Jugaad* is a way of survival for consumers at the base of pyramid (BoP) ([Prahalad, 2005](#)). When analyzing the definitions of the word “frugal” in the [dictionary.com](#) website, the results showed the following meanings: “economical in use or spending,” “prudently saving or sparing,” “requiring few resources” and “not wasteful.”

The concept of frugality was first inserted in the innovation field for the context of emerging markets, motivated by the necessity of meeting specific needs of the bottom of the pyramid consumers ([Prahalad, 2005](#)) from this type of market by providing solutions characterized by high added-value and low costs ([Radjou et al., 2012](#)). Inclusivity ([Ahuja, 2014](#)) is probably the main root of frugality, and it is expressed in different instances. In the previous study, [Micaëlli et al. \(2016\)](#) suggest another adaptation to integrate the different dimensions of frugality taking into account the inclusivity. According to the authors, market inclusivity, in the product level aims to target excluded clients, and in the process level considers clients as prosumers. The managerial inclusivity enables systems or solutions to promote the frugal innovation process for product level and implements frugal as agile and collaborative in the process level. The technical inclusivity improves the value of the product allowing user’s inclusivity through affordability, functionality, usability and agreement.

It is important to note that in the literature there are some innovation strategies associated with resource-constrained innovation because they have similar characteristics in common ([Bhatti, 2012](#); [Bhatti and Ventresca, 2013](#); [Zeschky et al., 2014](#)). These forms include *jugaad* innovation ([Radjou et al., 2012](#); [Prabhu and Jain, 2015](#)), Gandhian innovation

(Prahalad and Mashelkar, 2010), cost innovation (Zeschky *et al.*, 2014), good-enough innovation (Gadiesh *et al.*, 2007; Hang *et al.*, 2010) and reverse innovation (Govindarajan and Ramamurti, 2011).

Cost innovations are low-cost products, with cost reductions realized through process innovations and cost advantages in emerging markets. Good-enough innovations are tailored to the resource-constrained market, with non-value-adding functions eliminated and specific value-adding functions designed to meet the specific requirements of customers. Frugal innovations feature new applications developed specifically for resource-constrained environments, generating an entirely new value proposition. Finally, reverse innovations are cost, well-enough or frugal innovations that are transferred from emerging markets to developed-countries (Zeschky *et al.*, 2014, p. 24). Therefore, in our understanding, it seems that frugal innovation is more appropriate to explain the phenomenon about how to perform innovation under resource-constrained conditions than these other terms. Figure 1 depicts the types of resource-constrained innovations.

3. Methodology

This study combined two different methods to provide a good analysis and perspectives about the frugal innovation theme. In this sense, the SLR was performed by analyzing the literature on frugal innovation using as reference points the clusters found through the co-citation analysis. As the first objective was to understand the commonalities and differences in the scientific production that used the term frugal innovation to have a better comprehension of what it exactly means, this study followed the procedures of planning, execution and reporting (Tranfield *et al.*, 2003).

The planning stage is when the specific aims of the research are defined and the data sources are chosen. We decided to use Web of Science database as research source. According to Podsakoff *et al.* (2005), articles in this database are more likely to have the highest impact in the field of social sciences. “Of the journals in their database, approximately 5,500 of the journals are from the “hard” sciences, while the remaining 3,000 journals are from the social sciences and humanities” (Podsakoff *et al.*, 2005, p. 475). In

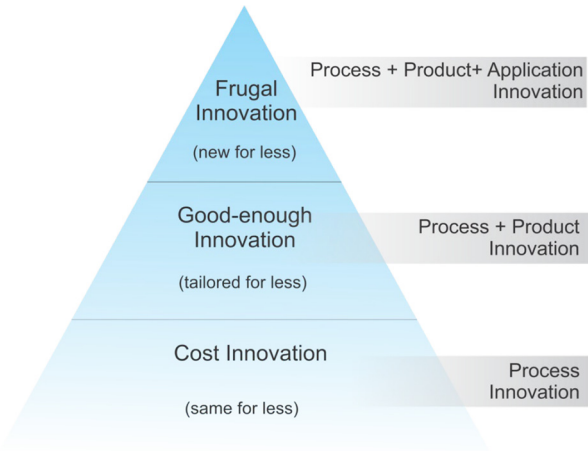


Figure 1.
Hierarchy of
resource-constrained
innovation

Source: Authors (2019). Adapted from Zeschky *et al.* (2014)

addition, this database was chosen because it provides specific metadata for the bibliometric analysis, including summaries, references, the number of citations, the list of authors and keywords.

The execution stage comprehends two steps as follows: data collection and data analysis. Data collection consists of the extraction of all articles from the scientific databases, started with the bibliographical research, which seeks to deepen the theoretical knowledge about the phenomena to be studied. In the data analysis, we then began to analyze the available documents to construct a set of papers for a full reading process. The first searching procedure brought 55 documents published between 2011 and 2019. The number of documents by year of publication are, respectively: 3 (2019); 18 (2018); 11 (2017); 14 (2016); 3 (2015); 2 (2014); 1 (2013); 2 (2012); and 1 (2011).

However, after analyzing this material we excluded 13 documents because they were just editorial materials (6), an interview (1), book reviews (2), reviews (3) and the other (1) was just a meeting abstract. So, in the selection step, we have chosen to keep with only papers and proceedings papers for the analysis. So, the final sample comprised a total of 42 documents selected from the Web of Science database. The selection of the papers followed some criteria such as:

- The first filter consisted of the time period of the publications, where the collection brought papers published between 2011 and 2019. We considered eight years as acceptable to address the existing literature so far because this study was performed in 2019 to map recent perspectives regarding this subject, and the first high cited paper is dated from 2011 ([Zeschky et al., 2011](#)), with 108 citations; and
- The choice of the term to be searched was based on publications using the term “frugal innovation” specifically in the filter “title” to avoid misconceptions. Another reason for this choice is because the term frugal innovation is the same chosen by [Zeschky et al. \(2011\)](#) and also by [Radjou et al. \(2012\)](#), who are the first authors according to the Web of Science metadata.

Finally, the reporting stage encompasses two analyzes as follows: the bibliometric network and the qualitative analysis of the clusters. According to [Van Eck and Waltman \(2014, p. 1\)](#), “the idea of visualizing bibliometric networks, also called as “science mapping,” has attracted attention, as scholars started to develop bibliometric research.” Visualization has become:

[...] a powerful approach to analyze a large variety of bibliometric networks, ranging from networks of citation relations between publications or journals to networks of co-authorship relations between researchers, or networks of co-occurrence relations between keywords ([Van Eck and Waltman, 2014, p. 1](#)).

To answer our research question, we performed a bibliometric technique called co-citation analysis, which connects references based on how they are cited together. While the number of citations increases, the influence of the references is also increased. The use of co-citation to analyze relations between researchers and between journals was first introduced by, respectively, [White and Griffith \(1981\)](#) and [McCain \(1991\)](#). Therefore, co-citation analysis reflects the past, that is, what was cited. The settings established to obtain the final network are: co-citation analysis as a type of analysis; cited authors as a unit of analysis; full counting as counting method; and 10 as a minimum number of citations of an author, which resulted in 16 authors to be selected, a number that we have not modified.

Finally, we performed a qualitative analysis of the selected papers to explain what is, where and to whom could be applied FIS. The qualitative analysis was performed through

in-depth reading of the 42 papers while we filled an Excel spreadsheet with important aspects from the papers such as research purpose, chosen approach, context addressed, main contribution and suggestion for future studies. Then, we analyzed all those details to clearly identify some standards such as the geographical context, consumer profile-related fields of study and other characteristics among the cited references (co-citation analysis), to highlight the recent perspectives from the selected papers.

4. Results

4.1 Co-citation analysis: the authors network analysis

In this section, the clusters identified by means of the co-citation analysis are presented (Figure 2). Therefore, it is analyzed the internal network of each cluster to establish the main theoretical foundations. Three clusters were created by the co-citation method. Cluster 1 is composed by seven authors (Agarwal, Bhatti, Hossain, Radjou, Soni, Tiwari and Zeschky). Cluster 2 is composed by four authors (Govindarajan, Hart, London and Prahalad). Finally, Cluster 3 have four authors as well (Bhatti, Brem, Rao and Mourtzis).

4.1.1 Cluster 1 – the focus of frugal innovation strategy. The first important aspect noticed is that the most cited authors are included in this cluster and they are specifically concerned about determining the boundaries of frugal innovation as a concept and its contexts of application. In the international context, the topic has been discussed as *jugaad* innovation in poor countries (Radjou *et al.*, 2013) and frugal innovation in rich nations (Radjou *et al.*, 2015). Radjou *et al.* (2013) state that *jugaad* innovation is based on specific principles originated from cases located in poor countries, which are manifested in expressions such as “doing more with less,” “aiming for simplicity” and “involving marginal populations.” On the other hand, Radjou *et al.* (2015) add other principles related to frugal innovation in rich countries such as “creating sustainable solutions,” “improving innovation process agility” and “shaping customers’ behavior.”

Agarwal *et al.* (2017) delimited three basic assumptions of constraint-based innovations as follows: cost-effectiveness, ease-of-use and prescriptive variables. In this sense, frugal innovation can have considerable consequences for development processes because of its focus on low cost and affordability, making life easier and improving the economic, social and environmental conditions of consumers (Agarwal and Brem, 2012).

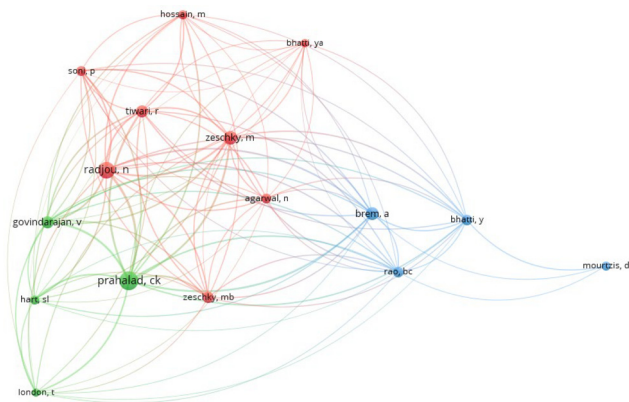


Figure 2.
Co-citation cluster
network

Source: Authors (2019). Output from Vosviewer software

Zeschky *et al.* (2011) assert that frugal innovation is a response to severe resource constraints by designing and developing products with extraordinary cost advantages if compared to existing solutions. In other words, frugal innovations are normally products or services developed for specific applicability by companies in resource-constrained environments. Furthermore, it is often quite disruptive (Zeschky *et al.*, 2014).

Bhatti (2012, p. 18) proposes that frugal innovation:

[...] may redefine business models, reconfigure value chains and redesign products to use resources in different ways and create more inclusive markets by serving users with affordability constraints, often in a scalable and sustainable manner.

In other words, frugal innovation involves (re)designing products, services or systems to significantly cut costs, without sacrificing user value, so as to reach a mass customer base, especially in low-income settings (Bhatti, 2012). In this sense, we can understand that frugal innovation is very linked to the affordability approach as it increases the importance of customer value for unattended demands.

Frugal innovation has been defined in different forms over the years. Although several definitions agree on some central features, the concept showed an evolution. In this cluster, we present the best definition we found in the literature so far in our opinion because it is the one, which seems to address the main characteristics identified by the authors in this cluster. Tiwari *et al.* (2014, p. 18) suggest that:

[...] frugal innovations are new or significantly improved products (both goods and services), processes, or marketing and organizational methods that seek to minimize the use of material and financial resources in the complete value chain (development, manufacturing, distribution, consumption, and disposal) with the objective of significantly reducing the total cost of ownership and/or usage while fulfilling or even exceeding certain pre-defined criteria of acceptable quality standards.

Soni and Krishnan's (2014) findings provided great contributions to the literature by highlighting that frugal innovations are of three types, namely, grassroots-level, domestic-enterprise level and multi-national-subsidiary level. This implies that this type of innovation occurs not only in economically different markets such as developed and emerging markets but also in different levels of geographical coverage such as local and global innovations.

Finally, the review performed by Hossain (2018) is a study that provides an overview of the ideas of the authors addressed in this cluster. Furthermore, in this study, the author brings back his definition of frugal innovation.

Frugal innovation as a resource scarce solution (i.e. product, service, process, or business model) that is designed and implemented despite financial, technological, material or other resource constraints, whereby the final outcome is significantly cheaper than competitive offerings and is good enough to meet the basic needs of customers who would otherwise remain underserved (Hossain *et al.*, 2016, p. 133).

According to the conceptualizations found in the analysis of this cluster, in this paper, we understand that frugal innovation is a low-cost technological solution, focused on the high competitiveness of the core functionality and oriented to a resource-constrained situation. Therefore, FIS is the one that has a purpose to achieve the frugal innovation outcome.

4.1.2 Cluster 2 – the Scopu of frugal innovation strategy. The concept of frugal innovation has stood out because of the experts' understanding that frugality requires a change in the mindset of companies operating in both emerging and developed markets. As frugal innovation has been considered as an alternative to meet the needs of populations

with limited access to resources, it has attracted special interest from emerging economies because they are characterized by resource-constrained environments (Prahalad, 2005).

Prahalad and Hart (2002) suggest that there is an opportunity to profit from poor demands by treating them as self-respecting customers. However, it should be noted the unequal bargaining power between global companies and the BoP consumers located in poor or developing countries. Among the overlapping concepts, maybe the frugal innovation closest relation is with BoP innovation, although BoP innovation only focuses on poorer customers with low annual incomes (Hart and Christensen, 2002). On the other hand, frugal innovations are normally considered to be potentially disruptive and transformational, not only for emerging markets but also for developed markets (Immelt *et al.*, 2009). In addition to attracting consumers from developing markets in Asia, Africa and Latin America, frugal innovation has also attracted cost-conscious consumers in Europe and North America (Govindarajan and Ramamurti, 2011).

Innovative ideas and products have been increasingly developed by local companies in emerging countries and globally marketed (Govindarajan and Ramamurti, 2011). In this sense, what Christensen (1997) had characterized as disruptive innovation, i.e. products that offer superior value to the customer and low costs, has particularly been developed in corporations from China and India, which are growing exponentially because of this type of innovation. (London and Hart, 2004).

The analysis of the second cluster allows us to conceive that the above-mentioned frugal innovation is neither restricted to the BoP nor to underdeveloped and emerging countries. Frugal innovation can also be designed for developed countries and traditional markets. Thus, the FIS aimed at a frugal technological solution may have as its market scope both traditional markets and the bottom of the pyramid, as it can be implemented in underdeveloped, emerging and developed countries.

4.1.3 Cluster 3 – the purpose of frugal innovation strategy. In this cluster, authors present other studies that explore some aspects of frugal innovation in perspectives of diffusion to other contexts that can be related. There is a connection between frugal innovation and themes such as sustainability and affordability. The authors of this cluster indicate the importance of fostering frugal innovation to help social and environmental problems.

Mourtzis *et al.* (2016) show the same point of view when the author states that frugal innovation seeks to provide high-value and local market solutions considering the local requirements in a cost-efficient way. Besides, the combination of global production and distributed customer networks forms the basis of frugal innovation, which aims at exploiting the concept of intelligent use of resources, turning the related constraints into advantages and a driving force toward product innovation. In this sense, we see that frugal innovation can be part of a global value chain, as Bhatti (2012) states that this dynamics of frugal innovation can redefine business models and reconfigure value chains.

Brem and Wolfram (2014, p. 19) conceptualized frugal innovation as an:

[...] approach which focuses on the development, production, and product management of resource-saving products and services for people at the BoP by achieving a sufficient level of taxonomy and avoiding needless cost.

Moreover, Rao (2017) highlights the relevance of motivating advances in science and technology to reach the frugality in resource for correct usage. Therefore, frugal innovations are also becoming noticed in developed economies because of their lower costs and no-frills structure, also involving rethinking entire production processes and business models (Rao, 2013). Therefore, according to Rao (2013), companies will have to adopt the principles of

frugal-innovations to achieve profitability to meet both cost-conscious consumers and regulations for sustainable development.

Frugal innovation combines local ideas and knowledge with international expertise in inclusive value chains to develop low-cost products that provide high added-value (Bhatti, 2012) in an efficient way, for instance, to create significantly more value to the user while minimizing the use of resources by the manufacturer (Bhatti, 2012; Bhatti and Ventresca, 2013). In this case, we see that frugal innovation is also applied to the process efficiency approach because of its focus on process improvement through resource harnessing and exploitation.

The analysis of the third cluster allows us to conclude our conception of FIS. The cluster shows us that frugal innovation has a purpose of value creation that goes beyond economic value. In other words, it is the objective of frugal innovation to create environmental and social value as well. Thus, the FIS must have in its conception the creation of economic, social and environmental value.

Finally, the analysis of the three clusters allows us to establish:

[...] frugal innovation strategy is a low-cost technological solution, focused on the competitiveness of the core functionalities and oriented to a resource-constrained situation, whose scope of application can be for both emerging and developed countries, and whose proposition is to create economic, social and environmental value.

4.2 *The boundaries of frugal innovation strategy*

Considering our conceptualization of FIS, we established four parameters to clarify FIS, namely, why, what is, where and to whom.

4.2.1 Why frugal innovation strategy. The explanation of why we called FIS is based on two points. The link with some main theories of strategy and the value proposition of FIS. First, the FIS is aligned with some theories that are the basis of strategic thinking. Here, we will analyze based on the three main perspectives of strategy (Peng *et al.*, 2009) as follows: resource-based view, institutional-based view and industry-based view. As frugality is encouraged by resource-scarce environments, weaker institutional intermediaries and a higher tolerance for uncertainty, it is possible to make a linkage of the frugal innovation theme with theories such as resource-based view, industry-based and institutional theory (Soni and Krishnan, 2014). The institutional environment and specifically the market failures are the reasons for the resource-scarce environment. It is strategically to the company to read the threats or opportunities to answer in a better and fast way to the resource-scarce situation. The FIS could be a strategic response (Shibin *et al.*, 2018). The company is obliged to develop capabilities to work under resource-constraints to perform the FIS. It means new and differentiated capabilities when compared with the traditional view of developing capabilities with better resources. For Kuo (2016), the main features of frugal innovation are the focus on fundamental needs, sparingly use resources and avoid non-essential functions to simply design and make products or services affordable for most consumers, enabling many organizations to rethink and redesign their products or services. Finally, all the effort to develop the FIS could result in a better strategic positioning aligned to cost reduction and differentiation. The cost reduction came from the aim of minimum cost in the frugal innovation and differentiation from the better performance of the main functionality.

The second reason for an explanation about the FIS is the value proposition. The value proposition of the FIS is shared between economic, social and environmental. The economic aspect is the base of all development of frugal innovation when we explain the better

performance with the minimum cost. With the rationalization of the use of resources and raw materials (Radjou *et al.*, 2015) harnessed to a significantly more value generation (Bhatti, 2012; Bhatti and Ventresca, 2013), the frugal innovation helps to increase the company's profits and sales by reaching more people (Prabhu, 2017). However, the articles showed the evolution of frugal innovation to a social and environmental thinking performance in all value chain. It involves the ability to generate value impacting the business and social environment by minimizing the use of resources such as energy, capital and time while maximizing value for stakeholders (Radjou and Prabhu, 2014).

Frugal innovation had a significant impact on society because it seeks to solve pressing social problems because of a lack of resources through improvisation but generating revenue as well (Bhatti, 2012). In this sense, frugal innovation has the potential to improve a company's sustainability performance (Brem and Ivens, 2013; Khan, 2016). Many companies adopted engineering and economic sustainability in their processes as a core concept for their business models (Radjou *et al.*, 2015; Prabhu, 2017). For Prabhu (2017), companies are now questioning the number of resources and raw materials used in their processes, how products are designed and packaged and how to be engaged with consumers more environmentally conscious. The needs of consumers with restricted access to limited resources are disseminated throughout the value chain of the companies, from the obtaining of raw materials to product production and durability, as an attempt to provide accessibility and to add value to these consumers (Hyvärinen *et al.*, 2016). It is believed that the sustainability of frugal innovations arises mainly from no-frills design and simplicity that frugal solutions provide (Rao, 2013; Basu *et al.*, 2013; Brem and Ivens, 2013). The quality of the outcomes is argued to lead to lower resource consumption and, therefore, increased sustainability (Rao, 2013; Brem and Ivens, 2013; Thøgersen, 2018).

4.2.2 What is frugal innovation strategy? In this sense, FIS has a focus on innovations that are suitable for companies in resource-constrained environments, given its characteristics of substantial cost reduction, focus on essential functionality and sustainable resource use (Weyrauch and Herstatt, 2016; Tiwari *et al.*, 2016). Frugal innovations are characterized by innovative solutions from the marketing and technological perspective (Zeschky *et al.*, 2014). However, as it is a new concept, the current application of frugal innovation processes is still varied and some solutions are developed from scratch, while others are built on existing technologies (Agarwal and Brem, 2012; Rao, 2013; Zeschky *et al.*, 2011). In both cases, the solutions have as the purpose of meeting the specific needs in their target markets. So, if the local circumstances are known, as well as the requirements, desires and behavior of end-users, the expectations of the success of the frugal solutions are high (Agarwal and Brem, 2012; Zeschky *et al.*, 2011).

In this way, FIS can be intertwined with the four main innovation types, namely, product, process, marketing and organizational (Tiwari and Herstatt, 2012). In this sense, FIS challenges all the steps in the innovation cycle, from idea to business model, marketing and consumption (Radjou and Prabhu, 2014). Because of their emphasis on affordability, frugal innovation processes often require companies to rethink their current operating practices (Hyvärinen *et al.*, 2016). From the marketing perspective, companies have also struggled to remove non-essential functions from their products, simplify their service architecture or reduce their process costs to offer more affordable products and services to address neglected demands and make a profit (Kuo, 2016).

Therefore, according to Zeschky *et al.* (2011), low-cost of design and manufacturing, combined with low-cost of materials are pre-requisites for achieving affordability, and if combined with the simplicity of basic functionalities and minimal feature sets it will be

established the necessary quality for the frugal solutions. In this sense, if the business model changes to adapt to this purpose, then the organizational innovation tend to be a frugal innovation as well. [Bhatti \(2012, p. 18\)](#) proposes that frugal innovation:

[...] may redefine business models, reconfigure value chains and redesign products to use resources in different ways and create more inclusive markets by serving users with affordability constraints, often in a scalable and sustainable manner.

4.2.3 Where develop and apply frugal innovation strategy. Frugal innovations first started in the context of emerging markets. The main idea was to develop products and services that fit these markets' special needs and requirements and that were cheap enough to give to unmet customers opportunities for consumption ([Hart and Christensen, 2002](#); [Prahalad and Hart, 2002](#); [Soni and Krishnan, 2014](#)). On the other hand, frugal innovations have also found their way into developed markets ([The Economist, 2010](#)) and are often referred to as reverse innovation ([Immelt et al., 2009](#); [von Zedtwitz et al., 2015](#)).

The frugal mindset was created in emerging markets, especially India and China. India has been considered by some scholars as to the leading market for frugal innovation ([Tiwari and Herstatt, 2012](#)) while others believe that India's potential as a "laboratory for frugal innovations" is over-rated ([Prathap, 2014](#)). [Bhatti and Ventresca \(2013\)](#) built a conceptual model of frugal innovation in emerging markets. The authors pointed out three main challenges that can hinder or impede the movement or progress of frugal innovation in emerging markets, which are scarcity and resource constraints, institutional gaps or complexities and relatively low individual purchasing power.

Frugal innovation requires a process and product pattern of engineering-oriented to a local context ([Liu et al., 2016](#)). Besides, understanding the local context is crucial for the success of frugal innovations, as they focus on achieving a radical cost goal and ensuring suitability for the local environment at the same time ([Agarwal and Brem, 2012](#); [Zeschky et al., 2011](#)). However, as mentioned earlier, in addition to guidance for emerging market consumers in Asia, Latin America and Africa, frugal innovation has also attracted cost-conscious and environmental-conscious consumers in Europe and North America ([Govindarajan and Ramamurti, 2011](#); [Radjou and Prabhu, 2014](#); [Knorringer et al., 2016](#)).

Because of globalization, the worldwide economy became more interconnected and emerging markets started to attract the attention of developed economies and receive investments from them ([Kuo, 2016](#)). The companies realized that if some frugal innovations developed to emerging markets are adapted they could be implemented in a developed market. Then, the reverse innovation phenomenon possibly brings frugal innovation to developed markets. In addition, frugal innovation has attracted cost-conscious and environmental-conscious consumers in Europe and North America ([Govindarajan and Ramamurti, 2011](#)), and developed countries are also starting to face resource scarcity and low growth rates ([Radjou et al., 2015](#)).

4.2.4 To whom is frugal innovation strategy? Regarding the type of consumer, frugal innovations are necessarily concentrated on low-cost and affordable solutions that make customer's lives easier improving the economic, social and environmental conditions of populations aiming to provide considerable advances in development processes for further progress ([Agarwal and Brem, 2012](#); [Bhatti, 2012](#)). Frugal innovation is developed under severe resource constraints, and it involves products or services with high quality and affordable prices, even for customers with modest lifestyles. Frugal innovations are good-enough, affordable products made to satisfy the needs of resource-constrained consumers ([Zeschky et al., 2011](#)). When designed to or from developed markets, we can see that FIS is oriented to consumers concerned with innovations that prevent unnecessary sophistication

and focus on the environmental sustainability because of resource-saving mentality tendency (Khan, 2016; Pisoni *et al.*, 2018).

To both set of countries, from the firm perspective, a frugal solution has to be designed, produced and delivered to meet the needs of underserved consumers in constrained environments. This approach takes a bottom-up view of how stakeholders, particularly large and small firms, can interact with consumers bringing a transition to sustainable economic progress (Prabhu, 2017).

Some examples of sectoral developments where frugal innovation is needed are: manufacturing better and cheaper products; agriculture processes to solve the problem of food waste; in the automotive industry it assumes increasing the fuel efficiency; and for the energy sector, for instance, moving to a carbon-free makes a renewable energy that conducts to a frugal energy consumption (Prabhu, 2017). The author understands that for the sustainable growth of the global economy, a systemic transformation to a frugal economical and industrial model is needed. It will be required for consumers to change behaviors and patterns of consumption to become more cost and environmentally conscious.

In sum, FIS has been pointed out as a potential approach for meeting resource-constrained consumers in emerging and developing markets, as well as in the low-growth Western markets (Agarwal and Brem, 2012; Rao, 2013; Zeschky *et al.*, 2011; Zeschky *et al.*, 2014) where it has been recognized the notion of affordability, good (enough) quality and no-frills structure (Hyvärinen *et al.*, 2016).

5. Discussion

After the conceptualization of FIS and the delimitation of boundaries of FIS, in the present section, we will bring the reflections about the contribution to the literature and the practice (managers and society) by showing three assumptions to be tested and confirmed in future studies and a framework to guide companies in search of a FIS.

5.1 Contributions to the literature: the boundaries of frugal innovation strategy

Frugal innovation has been pointed out as a potential approach for meeting resource-constrained consumers in emerging and developing markets, as well as in the low-growth Western markets (Agarwal and Brem, 2012; Rao, 2013; Zeschky *et al.*, 2011; Zeschky *et al.*, 2014) where it has been recognized the notion of affordability, good (enough) quality and no-frills structure (Hyvärinen *et al.*, 2016). Frugal innovation is concerned about generating greater value for the demands, which firms choose to generate this value for while at the same time it is greatly concerned about sparing the resources used to generate greater value (Prabhu, 2017). So, if the local circumstances are known, as well as the requirements, desires and behavior of end-users, the expectation of success of the frugal solutions are high (Agarwal and Brem, 2012; Zeschky *et al.*, 2011).

Frugal innovation demands a process and product pattern of engineering-oriented to emerging markets (Liu *et al.*, 2016). However, frugal innovation has also attracted cost-conscious and environmental-conscious consumers in Europe and North America (Govindarajan and Ramamurti, 2011), and developed countries are also starting to face resource scarcity and low growth rates (Radjou *et al.*, 2015). Based on the geographical context addressed in the “where” parameter, we can point out a first assumption: *frugal innovation is appropriated for both emerging and developed markets*.

It is important to highlight that often some characteristics of cost innovation and good-enough innovation have been treated as frugal innovation in the literature. Zeschky *et al.* (2014) sought to clarify commonalities and differences between these three types of resource-constrained innovations using two variables, which they called market novelty and

technical novelty. The authors showed evidence of how frugal innovation developed products or services for specific solutions in resource-constrained contexts. In fact, frugal innovation shares some principles with cost innovation and good-enough innovation, but it does not mean that these features are basically frugal innovation.

Zeschky *et al.* (2014) argued that there is a relation of dependency between these types of resource-constrained innovations, where frugal innovation is on the top of the pyramid, good-enough innovation is on the middle and cost innovation is on the basis. For the authors, cost innovations are enabled through process innovations that reduce operational costs, while good-enough innovation combines cost-cutting process innovations with the removal of superfluous features that are not important to resource-constrained customers while improving or customizing the solutions with attributes demanded by the target markets. In this sense, a third assumption we make is: *frugal innovation is achieved after experience in the cost innovation and good-enough innovation stages.*

One last assumption we make is that frugal innovations can be developed in emerging markets and afterward adopted by developed markets. Previous studies have also addressed that frugal is associated with the concept of reverse innovation (Von Zedtwitz *et al.*, 2015), which is represented by the reverse flow of products or services originated in developing markets and destined to developed markets (Immelt *et al.*, 2009). In this sense, innovations designed for local markets are increasingly being marketed globally (Zeschky *et al.*, 2011).

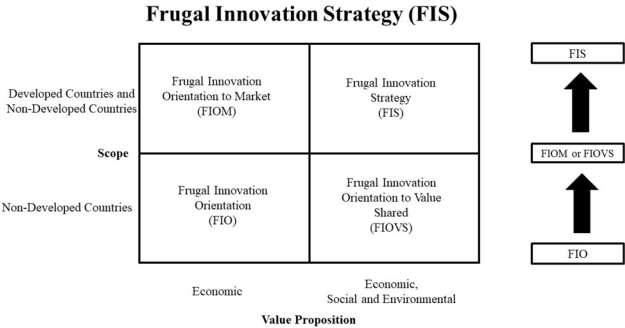
Agarwal and Brem (2012) distinguished frugal innovation and reverse innovation explaining that frugal innovation involves designing solutions specifically for low-income market segments, while reverse innovation involves new products developed in emerging markets, which are then modified for sale in developed countries. Zeschky *et al.* (2011) stated that frugal innovations have established large demands in emerging markets, and suggested that these innovations are developed by local research and development (R&D) subsidiaries of Western companies in emerging countries. The way for the success of frugal innovation depends also on the attention to promoting a mindset of frugality within R&D activities of subsidiaries in emerging markets, as well as to provide the necessary structure for the development of solutions for these contexts of restricted access to recourses. Therefore, the last assumption is: *frugal innovation can also be a reverse innovation.*

5.2 Contribution to the practice

Our conceptualization of FIS adds to the frugal innovation literature a new strategic orientation, which is useful for companies from emerging markets and developed markets (Govindarajan and Ramamurti, 2011; Radjou and Prabhu, 2014; Knorringer *et al.*, 2016). Although initially aimed at bottom-of-the-pyramid consumers, we have shown that the strategy can also be applied to above-base consumers in the consumer market (Bhatti, 2012; Tiwari and Herstatt, 2012; Rao, 2013; Prabhu, 2017). The FIS articulates the main theoretical streams of the strategy (Peng *et al.*, 2009). Furthermore, its objective is not only the company's economic sustainability but also social and environmental sustainability, in line with more current and broader perspectives of value creation in the strategy (Kramer and Porter, 2011). Finally, it is important to highlight that a FIS is not restricted to a certain kind of innovation, it encompasses different types of innovation (product, process, marketing and organizational), as the focus is on better performance, main functionality and low cost.

Once the FIS is conceptualized and has its boundaries delimited, the issues related to what it means, where and to whom it is applied allowed us to develop a framework for managers so they can establish their strategies to develop frugal innovations. Figure 3 shows our FIS model. In the framework, the y-axis stems from our delimitation of where and to whom the FIS is developed. On the one hand, the company may target the

Figure 3.
FIS framework



Source: Authors

underdeveloped market, usually as an attempt to serve the bottom of the pyramid or consumers unattended by market current solutions. On the other hand, the company may target not only this type of consumer but also consumers in the developed market. In turn, the *x*-axis arises from the strategic objective of the FIS that initially was characterized by the creation of economic value, evolving to the inclusion of social and environmental value.

The present framework configuration allows classifying the company in four strategic positions. However, before classifying it by scope and value proposition it is important that the company, in fact, has a focus on frugal innovation. Therefore, the first assessment is whether the company is developing an innovation strategy focused on the best performance of the core functionality in line with the lowest market cost for its development. This condition comes from our delimitation about what a FIS is, which has said previously, it can be geared toward the development of any kind of innovation, such as product, process, marketing, organizational and so on.

Indeed, if the company is developing a strategy oriented to frugal innovation, as mentioned above, then we can classify it in one of the four quadrants. A company has a frugal innovation orientation (FIO) when its main scope for innovation is a non-developed market, and when the value proposition of the technological solution is limited to a financial issue. We understand that this is the first stage in the search for the FIS. Note that besides our matrix we present FIO as the initial stage or the basic level in the path toward the FIS.

Following the framework, if the company already has a FIO, from an evolutionary learning perspective of technological development (Figueiredo, 2001), it can move to the upper quadrant named frugal innovation orientation to market (FIOM) or to the front strategic position named frugal innovation orientation to value shared (FIOVS). In the FIOVS strategy, the company maintains its scope focused on the underdeveloped market and designated for the initial consumer public to whom the frugal innovation is intended. However, the company in its strategic development now includes concerns about the creation of social and environmental value along with all stakeholders involved in the frugal technological solution. It is important to highlight that it is not just about the consumer/customer, but many frugal solutions involve strategic partners in the value chain and the impact of technology on society and the local environment as well. Therefore, this concern for shared value creation across the value chain becomes one of the main concerns for a company positioned in the FIOVS strategy.

In the FIOM strategy, the strategic purpose is still economic, but the company has as its scope not only the underdeveloped markets but also the developed markets become a target as

well. This kind of strategy requires the development of organizational capabilities to access this market and the ability to configure the technological frugal solution for markets with different institutional, competitive and consumer behavior features. In this case, the frugal technological solution must be capable of meeting different perspectives of consumer classes in markets with different stages of development. Regarding this peculiarity, we believe that in general companies that move from FIO to FIOM, do it through reverse innovation. In other words, the company develops a technological frugal solution for the underdeveloped market, which, in turn, a multi-national corporation sees potential for this innovation to be explored in developed markets in which it has units operating, to serve a very specific type of consumer of that market. The multi-national jointly with a local company, which may be a subsidiary or an offshoring operation, transfers the frugal solution and starts its commercialization in the developed market or allows it by third parties. The multi-national performs the commercialization when the innovation comes from a subsidiary. On the other hand, if the innovation comes from an external company (offshoring) the multi-national will have to negotiate whether it will be in charge of the commercialization or a partner that facilitates the commercialization of the frugal solution in the developed market.

In our framework, both FIOVS and FIOM strategies are at the intermediate level of frugal strategy development. The advanced level is characterized by the FIS itself. A company positioned at FIS has a FIS that serves both underdeveloped and developed markets, and it has a technological solution that creates economic, social and environmental value. The path toward FIS, considering our framework, can be through the evolution from FIOVS to FIS or from FIOM to FIS. In the first case, the company that already has a solution that creates shared value needs to develop the ability to explore new markets with that innovation. In the second case, the company already exploring its solution in different markets needs to add social and environmental value to the technological solution.

The comparison of the means to achieve the FIS deserves some attention. *A priori*, it could be taken into account, which one of the ways (FIO $\rightarrow$ FIOVS $\rightarrow$ FIS) or (FIO $\rightarrow$ FIOM $\rightarrow$ FIS) is the easiest. Each one has its own peculiarities and strategic challenges, and following the assumptions of technological learning (Figueiredo, 2001) both are at the intermediate level. That means even though the paths are different, the technological complexity of changing from one level to another is the same. Another point to be mentioned is the possibility of a strategy development skipping the intermediate level (FIO $\rightarrow$ FIS) or even going straight to FIS. We do not rule out the possibility, but it would be out of a technological learning perspective and in our view to be performed by few companies and with great risk of success because of the technological complexity that involves the strategic positioning of FIS.

Finally, it is important to highlight the impact of this framework for managers and society as well. The framework is useful for managers in two situations. Firstly, by positioning the company regarding the requirements of the FIS. When analyzing the framework the manager must address three questions. First, the manager must wonder about the company's innovation strategy, whether it is geared toward the best performance of core functionality and aligned with the lowest market cost. Otherwise, the company does not have a strategy for frugal innovation yet. Second, what is the market scope of the strategy, it is only for underdeveloped markets or it addresses developed markets as well. Third, what is the value proposition of the frugal innovation, economic or also social and environmental? Those three questions can help the manager to understand what is your level of technological complexity for frugal innovation and what is your strategy. The result can be level zero, that is, there is no strategy for frugal innovation; basic level, that is, there is a FIO; intermediate level, that is, the company has a FIOVS or FIOM strategy; or advanced level, when the company indeed has a FIS. Once identified with the company's positioning, the manager can be strategically prepared to set the

strategy to achieve the FIS. In case of already having a FIS, the manager can then understand how to keep the competitiveness of the company's FIS.

The impact on society lies both in public policy planning and the teaching of strategy and innovation. The FIS framework can guide both governmental, institutional and corporate investors in providing innovation funding. Depending on the aim of the investment fund and its purpose, the classification of the company's strategy in FIO, FIOVS, FIOM or FIS, can show investors, which company is a good target for the investment funding, and for the company, which investment fund it should apply for support. In turn, for the teaching of strategy and innovation at both undergraduate and postgraduate levels, our framework not only makes clearer what FIS is but also allows companies to be positioned to design strategies seeking the FIS.

6. Conclusion

The findings of this study provided by means of SLR a better view of frugal innovation through the co-citation analysis and the qualitative analysis, which were the basis for the parameters established. As prior discussed, frugal innovation has been addressed in different contexts and purposes where some authors considered to be convenient to their understanding. So, our intention was to organize the ideas about frugal innovation through the cluster analysis and the set of characteristics, which resulted in three parameters for the frugal innovation literature. Then, aiming to achieve this purpose it was performed a SLR by the analysis of 42 papers published since 2011.

The article has delimited the boundaries of FIS. The assumptions established in the discussion can become hypotheses for empirical studies. Specifically, what other scholars can have in mind is that frugal innovation is appropriated for both emerging and advanced markets; frugal innovation is more emphasized for product and process innovations; frugal innovation is achieved after experience in the cost innovation and good-enough innovation stages; and frugal innovation can also be reversed.

In addition, we explain why, what is, where and to whom the FIS can be developed and applied. Furthermore, we contribute by developing the FIS framework, with four strategic positions based on the boundaries of FIS organized by the complexity technological level. The strategic positions are FIO, FIOVS, FIOM and FIS. The effort of delimitation of the boundaries and the framework of FIS have an impact on literature, managers and society.

This research has some limitations. One limitation is the chosen database and the small sample used. Perhaps, a broader selection procedure considering other databases could show more documents that can contribute to our findings. Another limitation is the term chosen as a keyword for the data collection, as frugal innovation has been mistaken as a synonym of cost innovation and good-enough innovation, it is possible to have some documents addressing frugal innovation through these other designations. Although the use of the term can vary in each context, we believe we used the best one known, therefore, this does not jeopardize our research results but allow us to diminish the misconceptions instead.

Future research to aggregate advancements in the field should include analyzing different databases as a source, and other materials about the subject such as book chapters, working papers and managerial documents or reports. For future research, we encourage case studies to classify or show the evolution of the firms in the FIS framework.

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Further reading

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